5-MINUTE REVISION

Analysis of Variance

Partitioning total variation, degrees of freedom, mean squares, the F test, assumptions and ANOVA models.

Authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

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1 The idea

ANOVA (Analysis of Variance) compares population means by **partitioning the total variation** into parts due to known sources and to error.

Null hypothesis (one-way)

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_v$$

A significant F means **at least one pair** of means differs. It is an **omnibus** test and doesn't say which pairs.

2 Assumptions

- ✓ Observations from **normal** populations
- ✓ Common variance: **homogeneity of variance**
- ✓ Samples drawn **independently**
- ✓ Treatment effects are **additive**

So errors are iid $N(0, \sigma^2)$. Check with residual plots: residuals vs fitted values, and a normal Q-Q plot.

3 One-way ANOVA (CRD)

Correction factor

$$CF = \frac{G^2}{n}$$

Total SS

$$SS_T = \sum y_{ij}^2 - CF$$

Treatment SS

$$SS_{Tr} = \sum \frac{T_i^2}{n_i} - CF$$

Error SS

$$SS_E = SS_T - SS_{Tr}$$

Mean squares

$$MS = \frac{SS}{df}$$

F ratio

$$F = \frac{MS_{Tr}}{MSE}$$

v treatments, n observations, T_i treatment totals, G grand total.

4 Degrees of freedom

Source	One-way	Two-way
Treatments	$v-1$	$v-1$
Blocks	–	$r-1$
Error	$n-v$	$(v-1)(r-1)$
Total	$n-1$	$vr-1$

5 Two-way ANOVA (RBD)

Block SS (B_j block totals)

$$SS_B = \sum \frac{B_j^2}{v} - CF$$

Error SS

$$SS_E = SS_T - SS_{Tr} - SS_B$$

Separates treatment and block variation. With one observation per cell, interaction cannot be separated from error.

6 ANOVA models

Linear model

$$Y_{ij} = \mu + \tau_i + e_{ij}, \quad e_{ij} \sim N(0, \sigma^2)$$

Fixed	τ_i unknown constants (specific varieties of interest)
Random	$\tau_i \sim N(0, \sigma_\tau^2)$; variance components (randomly chosen fields)
Mixed	Both fixed and random effects

7 The ANOVA table

Source	df	SS	MS	F
Treatments	$v-1$	SS_{Tr}	$SS_{Tr}/(v-1)$	MS_{Tr}/MSE
Error	$n-v$	SS_E	MSE	
Total	$n-1$	SS_T		

Reject H_0 if calculated F > table F at ($v-1$, error df). Then use multiple comparisons to find which means differ.

! Common mistakes

- × Concluding **all** means differ from a significant F.
- × Forgetting to subtract the correction factor.
- × Deleting outliers just to change significance.

? Fill in the blanks

1. A CRD with 5 treatments and 4 replications has ——— error df. [15]
2. An RBD with 5 treatments in 4 blocks has ——— error df. [12]
3. MS(treatment) = 40 and MSE = 8 give F = ———. [5]
4. A grand total of 120 from 12 observations gives a correction factor of ———. [1200]
5. A treatment SS of 60 with 4 treatments gives a treatment mean square of ———. [20]
6. A significant F shows that at least ——— pair of means differs. [one]

Source: *Textbook of Agricultural Statistics*, Ch. 18 · Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul · KAU Me-LON
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